Algorithms for DNA Sequencing

Today's goal • 0/3 items

Module 1
DNA sequencing, strings and matching

Welcome

Welcome to Algorithms for DNA Sequencing
Reading • 10 min

Pre Course Survey
Reading • 10 min

Syllabus
Reading • 10 min

Setting up Python (and Jupyter)
Reading • 10 min

Getting slides and notebooks
Reading • 10 min

Using data files with Python programs
Reading • 10 min

Module 1: DNA sequencing, strings and matching

Module 1 Introduction
Video • 1 min

Lecture: Why study this?
Video • 4 min

Lecture: DNA sequencing past and present
Video • 3 min

Lecture: Genomes as strings, reads as substrings
Video • 5 min

Lecture: String definitions and Python examples
Video • 3 min

Practical: String basics
Video • 7 min

Practical: Manipulating DNA strings
Video • 7 min

Practical: Downloading and parsing a genome
Video • 6 min

Lecture: How DNA gets copied
Video • 3 min

Optional lecture: How second-generation sequencers work
Video • 7 min

Optional lecture: Sequencing errors and base qualities
Video • 6 min

Lecture: Sequencing reads in FASTQ format
Video • 4 min

Practical: Working with sequencing reads
Video • 11 min

Practical: Analyzing reads by position
Video • 6 min

Lecture: Sequencers give pieces to genomic puzzles
Video • 5 min

Lecture: Read alignment and why it's hard
Video • 3 min

Lecture: Naive exact matching
Video • 10 min

Practical: Matching artificial reads
Video • 6 min

Practical: Matching real reads
Video • 7 min

Quiz

Module 1
Graded Assignment • 30 min

Programming Homework

Programming Homework 1 Instructions (Read First)
Reading • 10 min

Programming Homework 1
Graded Assignment • 30 min

Welcome to Algorithms for DNA Sequencing

0:13 / 1:44 2x

Welcome to Algorithms for DNA Sequencing!

This module we begin our exploration of algorithms for analyzing DNA sequencing data. We'll discuss DNA sequencing technology, its past and present, and how it works. We'll learn more about DNA and genomes and we'll make the connection that a DNA molecule can be represented as a string of characters. This observation is the key that unlocks the entire rest of the course! It allows us to take methods from computer science that were invented to work with generic strings (web pages, books, files on your hard drive) and apply them to DNA.

We will also define the two computational problems that motivate our technical discussions for the rest of the course. The *read alignment* problem and the*assembly* problem. The rest of the course is divided into halves where we will discuss foundational algorithms and data structures for those two crucial problems.

In the practical sessions and homeworks, you will be working with real genomes and real sequencing data, including some human data. And you will start to write software to analyze them. You will learn how to work with important file formats like FASTA and FASTQ, and you will look for some basic patterns in that sequencing data. Finally, we will discuss the naive exact matching algorithm a simple, flexible, but not very fast algorithm for finding patterns in text. Next module we'll discuss algorithms that are a bit more clever, and see some new data structures as well.

Your first two assignments, Programming Homework 1 and the Module 1 Quiz are due this week. Please also be sure to read the syllabus for all of the details about course logistics.

Ben & Jacob

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